Chapter 2 — Force and Motion

Full textbook exercise · PECTAA Physics 11 · No ALP filtering

Q.1 — Multiple Choice Questions (with correct option and reason)

2.1. The angle at which dot product becomes equal to cross product magnitude is:

(a) 65° **(b)** 45°
(c) 75° **(d)** 30°

Answer: (b) 45° — cos θ = sin θ at 45°.

2.2. The projectile gains its maximum height at an angle of:

(a) 0° **(b)** 45°
(c) 60° **(d)** 90°

Answer: (d) 90° — $H = u^2 \sin^2\theta/2g$ is maximum at 90°.

2.3. The scalar product of two vectors is maximum if they are:

(a) perpendicular **(b)** parallel
(c) at 30° **(d)** at 45°

Answer: (b) parallel — cos 0° = 1.

2.4. The range of a projectile is same for two angles which are mutually:

(a) perpendicular **(b)** supplementary
(c) complementary **(d)** 270°

Answer: (c) complementary — θ and 90° − θ give the same sin 2θ.

2.5. The acceleration at the top of a trajectory of a projectile is:

(a) maximum **(b)** minimum
(c) zero **(d)** g

Answer: (d) g — Gravity acts throughout the flight.

2.6. SI unit of impulse is:

(a) kg m s^{-2} **(b)** N m
(c) N s **(d)** N s^{-1}

Answer: (c) N s — Impulse = force × time.

2.7. The rate of change of momentum is:

(a) force **(b)** impulse
(c) acceleration **(d)** power

Answer: (a) force — Newton's second law: $F = \Delta p/\Delta t$.

studysapiens.com

2.8. As a rocket moves upward during its journey, its acceleration goes on:

(a) increasing **(b)** decreasing
(c) remaining same **(d)** moving with uniform velocity

Answer: (a) increasing — Rocket mass decreases while thrust continues.

2.9. In an elastic collision there is:

(a) loss of energy **(b)** gain of energy
(c) no gain, no loss of energy **(d)** no relation between energy and elastic collision

Answer: (c) no gain, no loss of energy — Kinetic energy is conserved.

Q.2 — Short Answer Questions

2.1. State the right-hand rule for two vectors with reference to vector product.

Point the fingers of the right hand along A and curl them toward B through the smaller angle. The thumb gives the direction of A×B, perpendicular to the plane of A and B.

2.2. Define impulse and show how it is related to momentum.

Impulse is the product of average force and the time for which it acts: $J = F\Delta t$. From Newton's second law, $F = \Delta p/\Delta t$, so $F\Delta t = \Delta p$. Thus impulse equals the change in momentum.

2.3. Differentiate between an elastic and an inelastic collision.

In an elastic collision both momentum and kinetic energy are conserved. In an inelastic collision momentum is conserved but kinetic energy is not; some energy becomes heat, sound or deformation.

Feature	Elastic	Inelastic
Momentum	Conserved	Conserved
Kinetic energy	Conserved	Not conserved
Bodies	Separate	May stick together

2.4. Show that rate of change of momentum equals force. State Newton's second law in terms of momentum.

If momentum changes from p_1 to p_2 in time Δt, the rate of change is $(p_2 - p_1)/\Delta t$. Newton's second law states that this rate equals the applied force: $F = \Delta p/\Delta t$. For constant mass, $F = m\Delta v/\Delta t = ma$.

studysapiens.com

2.5. State the law of conservation of linear momentum and the condition under which it holds.

The total linear momentum of an isolated system remains constant if no external force acts on it. It follows from Newton's third law because internal impulses are equal and opposite.

2.6. Find the time of flight of a projectile to reach the maximum height.

At maximum height the vertical velocity is zero. Using $v_y = u \sin\theta - gt = 0$ gives $t = u \sin\theta/g$. This is half the total time of flight.

2.7. The maximum horizontal range of a projectile is 800 m. Find the height attained at 45°.

For maximum range, $\theta = 45°$ and $R_{max} = u^2/g = 800$ m. Height at 45° is $H = u^2 \sin^2 45°/(2g) = u^2/(4g) = R_{max}/4 = 200$ m.

2.8. Why does a hunter aiming directly at a bird in a tree miss the target?

The bullet follows a parabolic path and falls below the line of sight because of gravity. If the hunter aims exactly at the bird, the bullet drops during its flight and misses unless the bird also falls at the same instant.

2.9. Why does a person falling on sand not hurt as much as one falling on concrete?

Sand increases the time of impact. For the same change in momentum, a larger Δt gives a smaller average force, so the injury is less.

2.10. State the conditions under which birds fly in air.

A bird flies when lift balances its weight and thrust overcomes drag. The wing shape and angle of attack produce a pressure difference and downward deflection of air, giving upward lift.

2.11. Describe circumstances where velocity and acceleration are: (i) v zero but a not zero (ii) a zero but v not zero (iii) perpendicular.

(i) At the highest point of a vertically thrown body, $v = 0$ but $a = g$. (ii) In uniform straight-line motion, $a = 0$ but $v \neq 0$. (iii) In uniform circular motion, velocity is tangential and acceleration is radial, so they are perpendicular.

2.12. Describe briefly the effects of air resistance on range and maximum height of a projectile.

Air resistance opposes motion, reducing both horizontal and vertical velocity components. The range, maximum height and time of flight become smaller, and the path is no longer a perfect parabola.

studysapiens.com

Comprehensive / Long Questions

2.1 Define and explain scalar product. Write its important characteristics.

The scalar (dot) product of A and B is A·B = AB cos θ. It gives a scalar. It is commutative, distributive over addition, maximum for parallel vectors, zero for perpendicular vectors, and negative for obtuse angles. Work W = F·d is an example.

2.2 Define and explain vector product of two vectors. Discuss important characteristics.

The vector (cross) product is A×B = AB sin θ n̂, where n̂ is perpendicular to the plane of A and B by the right-hand rule. It is anti-commutative, distributive, zero for parallel vectors, maximum for perpendicular vectors, and its direction is normal to the plane. Torque τ = r×F is an example.

2.3 Derive the three equations of motion by graphical method.

For a velocity–time graph with uniform acceleration, slope gives a = (v − u)/t, so v = u + at. The area under the graph is a trapezium: s = ½(u + v)t. Substituting v gives s = ut + ½at². Eliminating t gives v² = u² + 2as.

2.4 What is projectile motion? Explain.

Projectile motion is the motion of a body thrown into the air that moves under gravity alone. It has constant horizontal velocity and vertically accelerated motion. The two motions are independent, and the path is a parabola when air resistance is neglected.

2.5 Derive expressions for (i) time of flight (ii) height attained (iii) range of a projectile.

(i) Setting vertical displacement to zero gives T = 2u sin θ/g. (ii) At the top v_y = 0, so H = u² sin²θ/(2g). (iii) R = u cos θ × T = u² sin 2θ/g.

2.6 Explain elastic collision in one dimension. Show that relative velocities before and after collision are equal in magnitude.

For masses m_1, m_2 with initial velocities u_1, u_2 and final velocities v_1, v_2, conservation of momentum and kinetic energy give $v_1 - v_2 = -(u_1 - u_2)$. Thus the relative speed of approach equals the relative speed of separation.

2.7 Explain elastic collision in two dimensions.

In two dimensions, momentum is conserved separately along x and y axes. Kinetic energy is also conserved. The final velocities are found by resolving momentum components and using the energy equation; the scattering angles depend on the impact parameter.

studysapiens.com

2.8 Explain an inelastic collision in one and two dimensions.

In an inelastic collision momentum is conserved but kinetic energy is not. In one dimension, if bodies stick together, their common velocity is found from total momentum. In two dimensions, momentum components are conserved along both axes, and the loss of kinetic energy appears as heat, sound and deformation.

Numerical Problems (fully worked)

2.1 The magnitudes of cross and scalar products of two vectors are 4√3 and 4. Find the angle.

Given: AB sin θ = 4√3, AB cos θ = 4

Formula: tan θ = (cross)/(scalar)

tan θ = 4√3/4 = √3, so θ = 60°.

Answer: 60°

2.2 A helicopter ascends at 19.6 m s⁻¹. At height 156.8 m a stone is dropped. How long to reach ground?

Given: $u = +19.6\ m\ s^{-1}$, $s = -156.8\ m$, $a = -9.8\ m\ s^{-2}$

Formula: $s = ut + \frac{1}{2}at^2$

$-156.8 = 19.6t - 4.9t^2 \rightarrow 4.9t^2 - 19.6t - 156.8 = 0$.

$t = [19.6 + \sqrt{(19.6^2 + 4\times4.9\times156.8)}]/(9.8) = 8.0\ s$.

Answer: 8.0 s

2.3 If |A+B| = |A−B|, prove A and B are perpendicular.

Squaring: $A^2 + B^2 + 2A\cdot B = A^2 + B^2 - 2A\cdot B$, so $4A\cdot B = 0$. Hence $A\cdot B = 0$ and θ = 90°.

Answer: θ = 90°

2.4 A body of mass M at rest explodes into 3 pieces; two of mass M/4 move perpendicularly at 3 and 4 m s⁻¹. Find velocity of third piece.

Given: $m_1 = m_2 = M/4$, $v_1 = 3$, $v_2 = 4$, $m_3 = M/2$

Formula: momentum conservation

Momenta of first two pieces: 0.75M and 1.0M, perpendicular. Resultant = 1.25M.

Third piece momentum = 1.25M opposite, so $v_3 = 1.25M/(M/2) = 2.5\ m\ s^{-1}$.

Answer: 2.5 m s⁻¹ opposite the resultant of the two pieces

The book prints 2.6 m s⁻¹; the momentum calculation gives 2.5 m s⁻¹.

2.5 A cricket ball is hit at 20 m s⁻¹ at 45°. Find (a) time of flight (b) maximum height (c) range.

Given: $u = 20\ m\ s^{-1}$, θ = 45°, $g = 9.8\ m\ s^{-2}$

Formula: $T = 2u\sin\theta/g$, $H = u^2\sin^2\theta/2g$, $R = u^2\sin2\theta/g$

studysapiens.com

T = 2×20×0.707/9.8 ≈ 2.9 s.

H = 400×0.5/(19.6) ≈ 10.2 m.

R = 400/9.8 ≈ 40.8 m.

Answer: (a) 2.9 s (b) 10.2 m (c) 40.8 m

2.6 A 20 g ball hits a squash wall with constant force 50 N for 0.50 s. Find impulse.

Given: F = 50 N, Δt = 0.50 s

Formula: J = FΔt

J = 50 × 0.50 = 25 N s.

Answer: 25 N s

2.7 A football experiences average force 240 N for 0.25 s. Find (a) change in momentum (b) direction.

Given: F = 240 N, Δt = 0.25 s

Formula: Δp = FΔt

Δp = 240 × 0.25 = 60 N s.

Direction is the direction of the force.

Answer: (a) 60 N s (b) in the direction of force

2.8 An aeroplane moves horizontally at 200 m s^{-1} at height 8 km. Find horizontal distance to release a bomb on target.

Given: $v_x = 200\ m\ s^{-1}$, h = 8000 m

Formula: t = √(2h/g), R = v_x t

t = √(16000/9.8) ≈ 40.4 s.

R = 200 × 40.4 ≈ 8080 m = 8.08 km.

Answer: 8.08 km

2.9 Why does range remain the same when θ changes to 90° − θ? Show R/R′ = 1.

R = u^2 sin 2θ/g. For θ′ = 90° − θ, sin 2θ′ = sin(180° − 2θ) = sin 2θ. Hence R′ = R and R/R′ = 1.

Answer: R/R′ = 1

2.10 A 1.0 kg trolley at 1.0 m s^{-1} collides with a similar trolley at rest. Show (i) elastic if first stops and second moves at 1.0 m s^{-1} (ii) inelastic if they stick at 0.5 m s^{-1}.

(i) Initial K = ½×1×1^2 = 0.5 J. Final K = 0.5 J, so elastic.

(ii) Common velocity = 1/(1+1) = 0.5 m s^{-1}. Final K = ½×2×0.5^2 = 0.25 J, so kinetic energy is lost and collision is inelastic.

studysapiens.com

Answer: (i) elastic (ii) inelastic

The book text mentions 0.6 m s^{-1}; momentum conservation gives 0.5 m s^{-1} for equal masses.

2.11 A railway wagon 4 × 10^4 kg at 3 m s^{-1} collides with 2 × 10^4 kg at rest and sticks. Find combined velocity.

Given: $m_1 = 4\times10^4$ kg, $u_1 = 3$, $m_2 = 2\times10^4$ kg, $u_2 = 0$

Formula: $v = (m_1u_1 + m_2u_2)/(m_1 + m_2)$

$v = (12\times10^4)/(6\times10^4) = 2$ m s^{-1}.

Answer: 2 m s^{-1}

2.12 A car 875 kg at 15.0 m s^{-1} hits a car 1575 kg at 5 m s^{-1} in same direction; they lock. Find (a) final velocity (b) K.E. lost.

Given: $m_1 = 875$ kg, $u_1 = 15$, $m_2 = 1575$ kg, $u_2 = 5$

Formula: momentum conservation; $\Delta K = K_i - K_f$

$v = (875\times15 + 1575\times5)/(2450) = 8.57$ m s^{-1}.

$K_i = ½\times875\times225 + ½\times1575\times25 = 118125$ J.

$K_f = ½\times2450\times8.57^2 \approx 90000$ J.

Loss ≈ 2.81×10^4 J.

Answer: (a) 8.57 m s^{-1} (b) 2.81 × 10^4 J

The book answer (7.67 m s^{-1}, 2.11 × 10^4 J) appears inconsistent with the printed masses and speeds; the values above follow from the given data.

Textbook checks flagged in this chapter

- Numerical 2.4: book prints 2.6 m s^{-1}; correct momentum result is 2.5 m s^{-1}.
- Numerical 2.10: book mentions 0.6 m s^{-1}; equal masses give 0.5 m s^{-1}.
- Numerical 2.12: book answer appears inconsistent with the printed data; solution uses the given masses and speeds.

studysapiens.com